

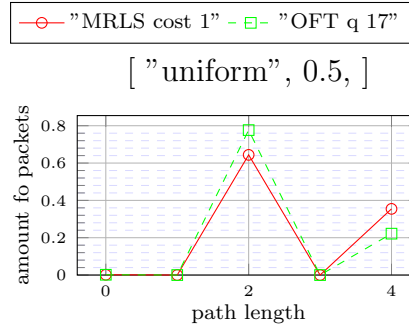


Figure 1: X: ["uniform", 0.5,]

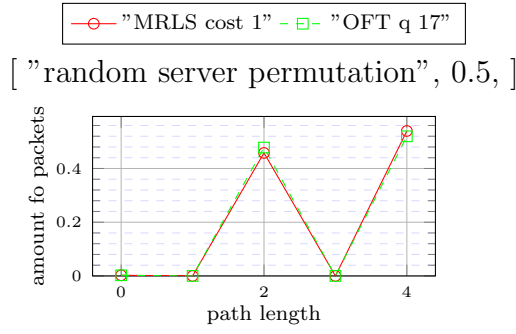


Figure 2: X: ["random server permutation", 0.5,]

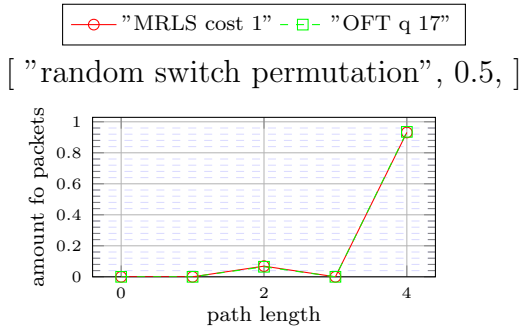


Figure 3: X: ["random switch permutation", 0.5,]

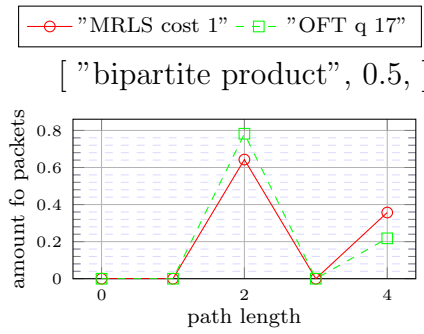


Figure 4: X: ["bipartite product", 0.5,]

The following versions used in the simulations.

- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  topology: ![
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_614_18_307_36" },
        servers_per_leaf: 18,
        legend_name: "MRLS cost 1"},
        OFT{ height: 1, prime: 17, servers_per_leaf: 18, legend_name: "OFT q 17" }},
    traffic: HomogeneousTraffic{
      pattern: ![
        Uniform{ legend_name: "uniform" },
        RandomPermutation{ legend_name: "random server permutation" },
        CartesianTransform{
          sides: [ 18, 614 ],
          patterns: [
            Identity,
            RandomPermutation{ seed: 500 }],
          legend_name: "random switch permutation"},
        Product{
          block_size: 5526,
          block_pattern: Uniform,
          global_pattern: CartesianTransform{
            sides: [ 2 ],
            shift: [ 1 ],
            legend_name: ""},
          legend_name: "bipartite product"}],
      servers: 11052,
      load: ![ 0.5 ],
      message_size: 16},
      maximum_packet_size: 16,
      router: InputOutput{
        virtual_channels: 4,
        virtual_channel_policies: [
          MapHop{
            hop_to_policy: [
              ArgumentVC{
                allowed: [ 0 ]},
              ArgumentVC{
                allowed: [ 0 ]},
              ArgumentVC{
                allowed: [ 1 ]},
              ArgumentVC{
                allowed: [ 1 ]},
              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 3 ]},
              ArgumentVC{
                allowed: [ 3 ]}],
            VecLabel{
              label_vector: [ 0, 64, 80 ]},
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
              use_neighbour_space: true, aggregate: true },
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
              use_neighbour_space: true, aggregate: false },
            LowestLabel,
            EnforceFlowControl,
            Random],
          allocator: Random,
          buffer_size: 128,
          bubble: false,
          flit_size: 16,
          intransit_priority: false,
          allow_request_busy_port: true,
          output_buffer_size: 64,
          crossbar_frequency_divisor: 1,
          crossbar_delay: 2},
        routing: ![
          Polarized{ include_labels: true, strong: false, legend_name: "Polarized" }],
        link_classes: [
          LinkClass{ delay: 2 },
          LinkClass{ delay: 2 }],
        launch_configurations: [
          Slurm{ job_pack_size: 1, time: "3-00:00:00" }]}]
```